

* $P = \prod_{k=2}^{\infty} \cos\left(\frac{\pi}{2^{k+2}}\right) = ?$

$P = \lim_{n \rightarrow \infty} P_n$ où $P_n = \prod_{k=2}^n \cos\left(\frac{\pi}{2^k}\right)$ où $U_n = \cos\left(\frac{\pi}{2^n}\right) \quad \forall n \in \mathbb{N}, n \geq 2$

$\forall n \in \mathbb{N}, n \geq 2$, posons $V_n = \sin\left(\frac{\pi}{2^n}\right)$ et $Q_n = \prod_{k=2}^n V_k$

On a: $P_n Q_n = \prod_{k=2}^n \left(\cos\left(\frac{\pi}{2^k}\right) \left(\sin\left(\frac{\pi}{2^k}\right) \right) \right)$

$= \prod_{k=2}^n \frac{1}{2} \sin\left(\frac{\pi}{2^{k-1}}\right)$ car $\forall x \in \mathbb{R}$, $\sin 2x = 2 \sin x \cos x$

$= \frac{1}{2^{n-2}} \times \sin\left(\frac{\pi}{2}\right) \times \left(\prod_{k=3}^{n-2} \sin\left(\frac{\pi}{2^{k-2}}\right) \right) \times \frac{1}{\sin\left(\frac{\pi}{2^n}\right)}$

$= \frac{1}{2^{n-2} \sin\left(\frac{\pi}{2^n}\right)} \times \prod_{k=2}^n \sin\left(\frac{\pi}{2^k}\right)$

d'où $P_n Q_n = \frac{1}{2^{n-2} \sin\left(\frac{\pi}{2^n}\right)} \times Q_n \Rightarrow P_n = \frac{1}{2^{n-2} \sin\left(\frac{\pi}{2^n}\right)}$ car $Q_n \neq 0 \quad \forall n \in \mathbb{N}, n \geq 2$

Ainsi $P = \lim_{n \rightarrow \infty} \frac{1}{2^{n-2} \sin\left(\frac{\pi}{2^n}\right)} \times \frac{2}{\pi} = \frac{2}{\pi}$ car pour $n = \frac{\pi}{2^n} \rightarrow 0$
 $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 1$

Donc $\prod_{k=2}^{\infty} \cos\left(\frac{\pi}{2^{k+2}}\right) = \frac{2}{\pi}$

* $Q = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2+\sqrt{2}}}{2} \times \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \times \dots = ?$

Suit $(u_n)_{n \geq 2}$: $\begin{cases} u_2 = \frac{\sqrt{2}}{2} \\ u_{n+1} = \frac{\sqrt{2+2u_n}}{2} = \sqrt{\frac{1+u_n}{2}} \end{cases}$

Montrons que $W_n = U_n \quad \forall n \in \mathbb{N}; n \geq 0$

- Pour $n=2$, $W_2 = \cos \frac{\pi}{2^2} = \frac{\sqrt{2}}{2} = U_2$

- Hérédité. $W_{n+2} = \sqrt{\frac{1+W_n}{2}}$
 $= \sqrt{\frac{1+\cos(\frac{\pi}{2^n})}{2}}$ car d'après l'hypothèse de récurrence $W_n = U_n \quad \forall n \in \mathbb{N}; n \geq 0$
 $= \sqrt{\cos^2(\frac{\pi}{2^{n+2}})}$
 $= |\cos \frac{\pi}{2^{n+2}}|$
 $= \cos \frac{\pi}{2^{n+2}}$ car $\forall n \in \mathbb{N}; n \geq 0, \frac{\pi}{2^{n+2}} \in]0; \frac{\pi}{2}[$

ainsi $W_{n+2} = U_{n+2}$

d'où $\cos \frac{\pi}{2^{n+2}} > 0$

CQFD

Par conséquent, car $Q = \lim_{n \rightarrow +\infty} \prod_{k=2}^n W_k = \lim_{n \rightarrow +\infty} \prod_{k=2}^n U_n = P$

alors $\boxed{Q = \frac{2}{\pi}}$

* De façon générale, écrivons, pour $\theta \in]0; \frac{\pi}{2}[$, les expressions
 $U_n = \cos \frac{\theta}{2^n}$, $V_n = \sin \frac{\theta}{2^n}$ ($n \in \mathbb{N}$), les expressions
 $P_n = \prod_{k=0}^n \cos \frac{\theta}{2^k}$, $Q_n = \prod_{k=0}^n \sin \frac{\theta}{2^k}$ et $P = \lim P_n$, $Q = \lim Q_n$.

$$\begin{aligned} P_n Q_n &= \prod_{k=0}^n \cos \frac{\theta}{2^k} \sin \frac{\theta}{2^k} = \prod_{k=0}^n \frac{1}{2} \sin \frac{\theta}{2^{k-1}} \\ &= \frac{1}{2^{n+1}} \times \left(\prod_{k=1}^{n+1} \sin \frac{\theta}{2^{k-1}} \right) \times \frac{\sin \theta}{\sin \frac{\theta}{2^n}} \end{aligned}$$

car ca $0 \in]0, \frac{\pi}{2}[$, $\frac{0}{2^n} \in]0, \frac{\pi}{2}[\Rightarrow \sin \frac{0}{2^n} \neq 0$

$$\text{d'où } P_n Q_n = \frac{\sin 2\theta}{2^{n+2} \sin \frac{\theta}{2^n}} \times \prod_{k=0}^n \sin \frac{\theta}{2^k} = \frac{\sin 2\theta}{2^{n+2} \sin \frac{\theta}{2^n}} \times Q_n$$

$$\text{d'où } \boxed{P_n = \frac{\sin 2\theta}{2^{n+2} \sin \frac{\theta}{2^n}}}$$

$$\bullet \text{ Ainsi } P = \lim_{n \rightarrow \infty} P_n = \frac{\sin 2\theta}{2\theta} \times \frac{\frac{\theta}{2^n}}{\sin \frac{\theta}{2^n}} \Rightarrow \boxed{\lim_{n \rightarrow \infty} P_n = \frac{\sin 2\theta}{2\theta}}$$

$\forall n \in \mathbb{N}$, car $\frac{\theta}{2^n} \in]0, \frac{\pi}{2}[$, $0 < V_n < \frac{\theta}{2^n}$

$$\text{d'où } 0 < Q_n < \prod_{k=0}^n \frac{\theta}{2^k} = \frac{\theta^{n+1}}{2^{\frac{n(n+1)}{2}}} = \left(\frac{\theta}{2^{\frac{1}{2}}}\right)^{n+1} = e^{\frac{(n+1) \ln \frac{\theta}{2^{\frac{1}{2}}}}{2^{\frac{1}{2}}}} \xrightarrow{n \rightarrow \infty} 0$$

$$\text{d'où } \underline{Q = \lim_{n \rightarrow \infty} Q_n = 0}$$

* Cas de $T = \lim_{n \rightarrow \infty} T_n$ où $T_n = \prod_{k=0}^n t_n$ où t_n change $\frac{\theta}{2^n}$ non

$$\forall n \in \mathbb{N}, t_n = \frac{V_n}{V_n} \text{ d'où } T_n = \frac{Q_n}{P_n} \xrightarrow{n \rightarrow \infty} \frac{0}{\frac{\sin 2\theta}{2\theta}} = 0$$

$$\text{Donc } \boxed{T = \lim_{n \rightarrow \infty} T_n = 0}$$

Soient $\theta_k, k=1, \dots, n \in \mathbb{R}^n$, $\cos(\sum_{k=1}^n \theta_k) = z ?$ / $\sin(\sum_{k=1}^n \theta_k) = z ?$

$$\cos \sum_{k=1}^n \theta_k = \operatorname{Re} \left(\prod_{k=1}^n e^{i\theta_k} \right) = \operatorname{Re} \left(\prod_{k=1}^n (\cos \theta_k + i \sin \theta_k) \right)$$

~~$\cos \sum_{k=1}^n \theta_k =$ on les termes réels de $P_n = \prod_{k=1}^n (\cos \theta_k + i \sin \theta_k)$~~

~~soit ceux pour lesquels~~ on

$$\prod_{k=1}^n (\cos \theta_k + i \sin \theta_k) = \sum_{\substack{J \subseteq \{1, \dots, n\} \\ |J| \text{ pair}}} \sum_{\substack{K \subseteq \{1, \dots, n\} \\ |K| \text{ pair}}} \left(\prod_{j \in J} (i \sin \theta_j) \right) \left(\prod_{k \in K} \cos \theta_k \right)$$

on $\operatorname{Im} P_n = \operatorname{Im} \left(\prod_{k=1}^n (\cos \theta_k + i \sin \theta_k) \right)$

$$\bar{J} = \{1, \dots, n\} \setminus J \quad \forall J \subseteq \{1, \dots, n\} \quad + \sum_{k=0}^{n/2} \sum_{\substack{J \subseteq \{1, \dots, n\} \\ |J| = 2k+1}} \prod_{j \in J} (i \sin \theta_j) \left(\prod_{j \in \bar{J}} \cos \theta_j \right)$$

on $\cos^2 \theta_k = \frac{1}{2}(1 + \cos 2\theta_k)$

$$\cos^{2k+1} \theta_k = \frac{1}{2} \cos^{2k} \theta_k \quad = \sum_{k=0}^{n/2} \cos^{2k} \theta_k \sum_{\substack{J \subseteq \{1, \dots, n\} \\ |J| = 2k}} \prod_{j \in J} \sin \theta_j \prod_{j \in \bar{J}} \cos \theta_j$$

$$+ i \left(\sum_{k=0}^{n/2} \cos^{2k} \theta_k \sum_{\substack{J \subseteq \{1, \dots, n\} \\ |J| = 2k+1}} \prod_{j \in J} \sin \theta_j \prod_{j \in \bar{J}} \cos \theta_j \right)$$

Car $\operatorname{Re}(P_n) = \cos(\sum_{k=1}^n \theta_k)$ et $\operatorname{Im}(P_n) = \sin(\sum_{k=1}^n \theta_k)$,

Alors $\cos \left(\sum_{k=1}^n \theta_k \right) = \sum_{k=0}^{n/2} \cos^{2k} \theta_k \sum_{\substack{J \subseteq \{1, \dots, n\} \\ |J| = 2k}} \prod_{j \in J} \sin \theta_j \prod_{j \in \bar{J}} \cos \theta_j$

et $\sin \left(\sum_{k=1}^n \theta_k \right) = \sum_{k=0}^{n/2} \cos^{2k} \theta_k \sum_{\substack{J \subseteq \{1, \dots, n\} \\ |J| = 2k+1}} \prod_{j \in J} \sin \theta_j \prod_{j \in \bar{J}} \cos \theta_j$

On particulier lorsque $\theta_k = 0 \forall k$ et $\forall k \in \{1, \dots, n\}$, on a:

$$\cos n\theta = \sum_{k=0}^{n/2} (-1)^k C_n^{2k} \sin^{2k} \theta \cos^{n-2k} \theta$$

$$\sin n\theta = \sum_{k=0}^{(n-1)/2} (-1)^k C_n^{2k+1} \sin^{2k+1} \theta \cos^{n-2k-1} \theta$$

Application:

$$\cos(\theta_1 + \theta_2 + \theta_3) = \cos \theta_1 \cos \theta_2 \cos \theta_3 - (\sin \theta_1 \sin \theta_2 \cos \theta_3 + \sin \theta_1 \sin \theta_3 \cos \theta_2 + \sin \theta_2 \sin \theta_3 \cos \theta_1)$$

$$\sin(\theta_1 + \theta_2 + \theta_3) = \sin \theta_1 \cos \theta_2 \cos \theta_3 + \sin \theta_2 \cos \theta_1 \cos \theta_3 + \sin \theta_3 \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \sin \theta_3$$

$$\cos(4\theta) = \cos^4 \theta - 6 \sin^2 \theta \cos^2 \theta + \sin^4 \theta$$

$$\sin(4\theta) = 4 \sin \theta \cos^3 \theta - 4 \sin^3 \theta \cos \theta$$

* Soit $n \in \mathbb{N}^*$, $\theta \in \mathbb{R}$, $\cos^n \theta = \sum ?$, $\sin^n \theta = \sum ?$

$$\cos^n \theta = \left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right)^n = \frac{1}{2^n} \left(\sum_{k=0}^{n/2} C_n^k e^{ik\theta} e^{i(n-k)\theta} + \sum_{k=0}^{n/2} C_n^k e^{-ik\theta} e^{-i(n-k)\theta} \right)$$

ou $C_n^{n/2} = 0$ pour $n \notin \mathbb{N}$

$$= \frac{1}{2^n} \sum_{k=0}^{n/2} C_n^k \left(e^{i(n-k)\theta} + e^{-i(n-k)\theta} \right) + C_n^{n/2}$$

Donc $\cos^n \theta = \frac{1}{2^{n-2}} \left(\left(\sum_{k=0}^{\frac{n-1}{2}} C_n^k \cos \theta (\ln 2k) \right) + \frac{C_n^{\frac{n}{2}}}{2} \right)$

De même

$$\sin^n \theta = \frac{1}{2^{n-2}} \left(\sum_{k=0}^{\frac{n-1}{2}} C_n^k e^{-i\theta \ln 2k} \times (i)^k + C_n^{n-k} e^{i\theta \ln 2k} \times (i)^{n-k} + \frac{(-1)^{\frac{n}{2}} C_n^{\frac{n}{2}}}{2} \right)$$

• Si $n \equiv 0 \pmod{2}$, $\sin^n \theta = \frac{(i)^{\frac{n}{2}}}{2^{n-2}} \left(\frac{C_n^{\frac{n}{2}}}{2} + \sum_{k=0}^{\frac{n-2}{2}} C_n^k (i)^k \cos \theta (\ln 2k) \right)$ " pour $n \equiv 1 \pmod{2}$

• Si $n \equiv 1 \pmod{2}$, $\sin^n \theta = \frac{(i)^{\frac{n+1}{2}}}{2^{n-2}} \sum_{k=0}^{\frac{n-1}{2}} C_n^k (i)^k \sin \theta (\ln 2k)$

Application :

$$\cos^2 \theta = \frac{1}{2} (\cos 2\theta + 1), \quad \cos \theta = \frac{1}{2^0} (\cos \theta)$$

$$\cos^3 \theta = \frac{1}{4} (\cos 3\theta + 3 \cos \theta)$$

$$\sin \theta = \frac{(i)^0}{2^0} (\sin \theta), \quad \sin^2 \theta = \frac{(i)^1}{2} (1 + \cos 2\theta) = \frac{1 - \cos 2\theta}{2}$$

$$\sin^3 \theta = \frac{-1}{4} (\sin 3\theta + 3 \sin \theta) = \frac{1}{4} (3 \sin \theta - \sin 3\theta)$$

Commentaires : Dans l'ensemble

$$\sum_{k=k_1}^{k_2} f(k) = \sum_{\substack{k_1 \leq k \leq k_2 \\ k \in \mathbb{N}}} f(k)$$

• $C_n^{y_2}$ a des prise pour 0 pour $n \in \mathbb{N}$